Orbital Advection Across Azimuthal Level Jumps in the AthenaK Shearing Box

Validation Document

`athenak-multigrid` tree, branch `fargo-amr`
parent commit `d3edb948` (branch `dust-multigrid`)

October 28, 2026

Abstract

Orbital advection (FARGO) in the AthenaK shearing box now accepts x_2 neighbors one level coarser or finer, so refinement may cover only part of the azimuth (method: `fargo_amr_implementation.pdf`). This document records the tests. Every annular and uniform configuration of the earlier test suite, hydro and MHD, is bit-identical to the parent code. With the orbital-advection flux correction the remap conserves mass to round-off across static and moving level jumps ($\sim 5 \times 10^{-14}$ over hundreds of steps; 10^{-7}–10^{-6} without it). Serial and 3-, 4- and 7-rank runs are bit-identical. A spatially uniform epicycle stays exactly uniform on a refined patch. Vortical shearing waves on a static patch and under a moving box lie between the ring and coarse-grid results, with no visible artifact at the level jumps. Swing amplification with multigrid self-gravity on a patch moves the peak 0.60% toward linear theory, and a gravity-off control shows that gravity adds no deviation of its own. A timestep limit lets a run proceed where the shift would otherwise exceed the neighbor's rows. For MHD (branch `fargo-amr-mhd`) orbital advection now runs on refined meshes, rings and partial-x_2 patches alike: the remap's EMFs are corrected at level jumps, which keeps the two copies of every shared face equal to round-off (10^{-2} apart without it), and the refined-mesh field error of the magnetized swing (13–22% without orbital advection) is gone: refined runs follow their uniform twins to 1–3% and linear theory better than the uniform runs. A pre-existing $\nabla \cdot B$ defect of AthenaK's AMR derefinement (10^{-13}–10^{-12} per event next to fine x_2 neighbours) is documented. The production test — SC14 clump-following AMR with and without rings — is still to be run.

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1 Test ladder

All runs are on a Mac (Apple Silicon, serial build and Open MPI build), October 28, 2026. Run directories are in `validation/run/fargo_amr/`; its `README.md` lists the command for each one. "Patch" means a refined region that covers only part of x_2, so it has x_2 level jumps; "ring" means the old annular configuration.

Test	What it probes	Result
Regression (§2)	same-level path unchanged	bit-identical
Epicycle on a patch (§3)	buffers, indices, prolongation	exact
Conservation (§4)	flux correction, static and AMR	5×10^{-14}
MPI (§5)	per-sub-face messages and tags	bit-identical
Vortical shearing wave (§6)	structure through level jumps	between ring and coarse
Swing with gravity (§7)	multigrid at x_2 level jumps	+0.60% toward theory
Timestep limit (§8)	large shifts	runs, conserves
MHD on rings (§9)	orbital EMF correction, field error	faces to 10^{-15}; 1–3%
MHD across x_2 jumps (§9.5)	B_1, B_3 rows across levels	faces to 10^{-15}; 1–3%
SC14 production (§10)	cost	not yet run

2 Regression: nothing changes where nothing jumps

On a same-level face the new code performs the same arithmetic as before, only with a different buffer layout and depth. Each input below was run with a build of the parent commit and with the new build, and the binary dumps (the data after the parameter header, which differs by the added input line) and history files compared byte by byte. All are **bit-identical**:

Input (`inputs/shearing_box/`)	Configuration
`epicycle_smr_ring`	level-1 ring, 1 orbit
`epicycle_amr_ring`, `epicycle_amr_ring2`	moving ring, 1 and 2 levels, incl. block create/destroy sequence
`hydro_incompress_shwave_smr_ring{,2}`	ring, 1 and 2 levels
`hydro_incompress_shwave_amr_ring`	moving ring
`epicycle`, `hydro_incompress_shwave`, `hydro_compress_shwave`	uniform mesh (VTK dumps for the shwaves)
`mhd_compress_shwave`	uniform mesh, MHD with orbital advection

The AMR ring inputs were run with `oa_annular = true`, which they need for the ring-wise flag synchronization (now off by default). The new orbital-advection timestep limit does not bind in any of them.

3 The epicycle on a patch

`epicycle_smr_ypatch.athinput`: $20 \times 20 \times 1$ box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks; the level-1 region covers $|x|, |y| < 5$, two of the four block rows in y, so four x_2 faces are level jumps. The `ipert=1` epicycle is spatially uniform at all times, so any error in buffer layout, indexing or prolongation appears as $O(1)$ structure. After one orbit (365 steps):

- density, all velocity components and the internal energy are uniform to the last digit of the single-precision output in every one of the 44 blocks;

- every history column matches the ring run (`epicycle_smr_ring`), and the mass does not change;
- on 4 MPI ranks every dump is bit-identical to the serial run.

4 Exact conservation

A uniform field cannot show a conservation error, and the vortical shwave of §6 has amplitude 10^{-4} and constant density. The conservation test is therefore the compressible shwave (`hydro_compress_shwave_smr`, isothermal, HLLE, PLM) at amplitude 0.1 on a 4×4 box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks, with a level-1 patch over $|x|, |y| < 1$. The x_1 boundaries are plain *periodic*: the shear-periodic remap at the x_1 faces is itself conservative only to $\sim 10^{-8}$ and would hide the effect. Only the hydro update and the orbital remap then move mass. The moving-box variant uses the AMR criterion of the implementation document with $x_c = 0.9 \sin t$, $y_c = 0.8 \sin 2t + t$ and half-widths 0.3.

maximum $\lvert M(t) - M(0)\rvert / M(0)$	flux correction on	off
static patch, $t = 0$–2 (453 steps)	5.2×10^{-14}	1.8×10^{-7}
moving box (AMR), $t = 0$–4 (899 steps; 98 blocks created, 84 destroyed)	5.7×10^{-14}	1.0×10^{-6}
static patch, shear-periodic x_1	3.8×10^{-9}	1.1×10^{-7}

With the correction the remap is conservative to round-off, through regridding too (Fig. 1); the total y-momentum stays at 3×10^{-16} against 5×10^{-7} without. With shear-periodic x_1 boundaries the correction still removes most of the drift; the remaining 4×10^{-9} is the boundary remap.

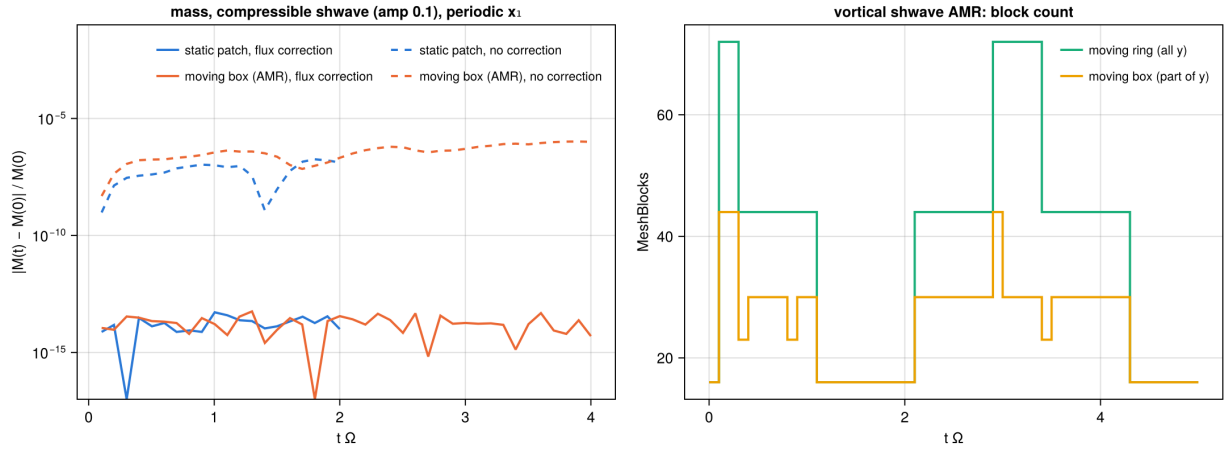


Figure 1: Left: relative mass drift of the compressible shwave (amplitude 0.1, periodic x_1) on a static patch (blue) and under a moving box with AMR (orange), with (solid) and without (dashed) the orbital-advection flux correction. Right: MeshBlock count of the vortical-shwave AMR runs of §6 — the moving box refines part of the ring's blocks. Script `figures/plot_fargo_amr_tests.jl`.

5 MPI

The conservation runs (static patch and moving box) were repeated on 3, 4 and 7 ranks. All binary dumps are bit-identical to the serial runs. The history files differ only in the volume sums, by the MPI reduction order: $\leq 6.5 \times 10^{-13}$ on a mass of 16, and $\sim 10^{-15}$ in momenta whose value is itself

round-off. Mass is conserved to 1×10^{-14} on every rank count. The vortical-shwave moving box on 4 ranks is likewise bit-identical to the serial run in all 51 dumps.

6 Vortical shearing wave through static and moving level jumps

The Johnson & Gammie (2005) vortical shwave (`ipert=2`, isothermal, PPM; $64^2 \times 4$ root cells on 0.5×0.5, $16^2 \times 4$ blocks, $t = 0$–5) on five meshes, compared with a uniform 128^2 run as the reference:

run	refined region (level 1)	peak x-KE vs 128^2	max $\lvert\Delta \mathrm{KE}_x\rvert$/peak
uniform 64^2	—	-4.5%	1.1×10^{-1}
ring, static	$\lvert x\rvert < 0.125$, all y	-7.9%	8.2×10^{-2}
patch, static	$\lvert x\rvert, \lvert y\rvert < 0.125$	-8.1%	9.3×10^{-2}
moving ring, AMR	$x_c = 0.2 \sin t$	-12.2%	1.3×10^{-1}
moving box, AMR	$+\, y_c = 0.15 \sin t + 0.1t$	-11.9%	1.3×10^{-1}

Inputs: `hydro_incompress_shwave{,_smr_ring,_smr_ypatch,_amr_ring,_amr_box}`; all history output at full precision (`output1/data_format=%.15e`). The patch lies between the ring and the coarse grid in the history error, and the moving box matches the moving ring (the box creates and destroys 77 blocks, the ring 112; Fig. 1, right). The peak error of the ring is larger than that of the coarse grid; this is a property of the ring runs, not of this work.

Figure 2 maps the v_x error against the 128^2 run. Inside the static patch the error is larger than inside the ring at $t = 1$: the wave reaches the fine region after crossing coarse cells in y, while in the ring it never leaves the fine grid. At the x_2 level jumps the error bands run straight across, with no stripes or kinks. Averaged over the refined x_1 columns of the patch, the error next to the jumps lies between those of the fine interior and the coarse rows ($t = 1$: 9.1, 4.9 and 11×10^{-2} of max $\lvert v_x\rvert$; at $t = 2.5$ the three agree to 10%).

The planned twin without orbital advection is not available for patches: with `orbital_advection = false`, the patch mesh collapses at the first step ($\Delta t \to 10^{-45}$) *in the parent code as well*, with or without `prolong_primitives`. The ring mesh runs. This is a separate, pre-existing problem (§10).

7 Swing amplification with multigrid gravity on a patch

`tests/swing_selfgrav_smr_ypatch.athinput` is the swing test of `multigrid_sb_validation.pdf` (Phase 2a: $2\pi \times 2\pi \times \pi/4$, $128^2 \times 16$ root cells, 16^3 blocks, $4\pi G\rho_0 = 1.9$, a leading wave with $k_x/k_y = -6$) with the level-1 region cut to the middle four of eight block rows in y (six of eight columns in x, as for the ring). The x_2 faces at $y = \pm\pi/2$ are then level jumps for orbital advection and for the multigrid octets. The ring and uniform runs were repeated with the new code.

run	peak $d_{\cos}/A$	peak time	max $\lvert d_{\sin}\rvert/A$
uniform, multigrid	-3.511418	7.804	7.3×10^{-14}
ring	-3.583481	7.804	1.3×10^{-13}
patch	-3.532422	7.804	6.1×10^{-13}

The uniform and ring runs reproduce the Phase-2a values to the last digit. The patch moves the peak 0.60% toward the converged answer (linear theory 3.605), under a third of the ring's 2.05%, with half the ring's refined area; the phase lock stays at machine level. The gravity-off control (`hydro_srcterms/self_gravity=false`) attributes the shift as in Phase 2a: the patch-vs-uniform deviation is larger without gravity (5.8×10^{-2} maximum, peak +1.11%) than with it (3.6×10^{-2}, +0.60%), so multigrid gravity across the x_2 level jumps adds no deviation of its own (Fig. 3).

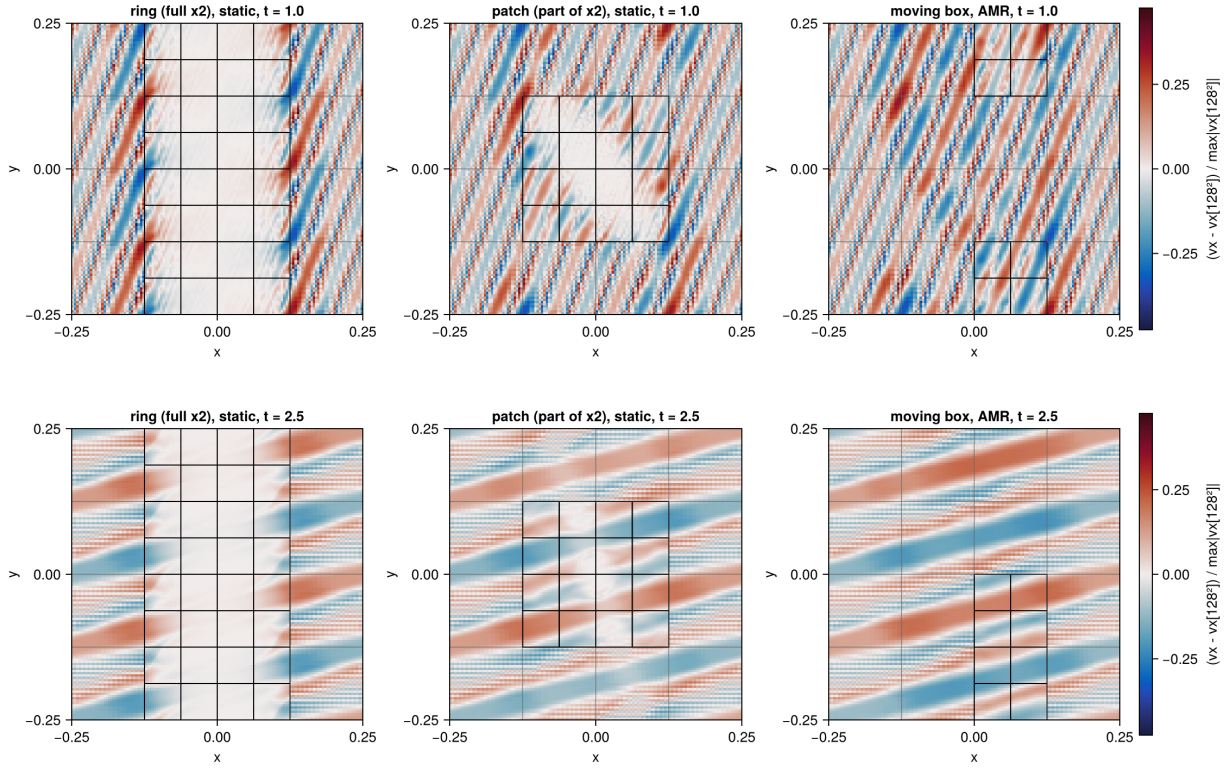


Figure 2: v_x error of the vortical shwave against a uniform 128^2 run, normalized to $\max|v_x|$, averaged over z, shown at the finest resolution (coarse cells repeated), at $t = 1$ (top) and $t = 2.5$ (bottom). Left: ring. Middle: static patch, x_2 level jumps at $y = \pm 0.125$. Right: moving box (AMR). Black: refined blocks; grey: root blocks. The checkerboard in coarse regions is the coarse cells against the fine reference. Script `figures/plot_fargo_amr_shwave.jl`.

8 The timestep limit

A cold variant of the static-patch conservation test (sound speed 0.05, amplitude 10^{-3}) lets the hydro timestep grow until the shift at $|x| = 2$ would be 8 cells against 6 available rows. The build without the limit stops at the first step with the guard's message. With it, Δt is held at $0.999\,(6-2)\,\Delta x_2/(q\Omega \max|x|) = 0.0416$, exactly the predicted value; the run reaches $t = 8$ (193 steps) and conserves mass to 8×10^{-14}.

9 MHD with orbital advection on refined meshes

Branch `fargo-amr-mhd`, in two steps. First, MHD with orbital advection on refined meshes whose refinement comes in complete x_2 rings (`oa_annular = true`; the subsections up to the magnetized swing); then across x_2 level jumps (§9.5). The effective EMFs of the face-field remap are corrected at level jumps like the stage EMFs (implementation document, section "Magnetic fields: MHD on rings"). Before this, MHD + orbital advection + refinement was a fatal error, and the refined MHD shearing boxes of `multigrid_sb_validation.pdf` (Phase 4) ran without orbital advection, where the field perturbation of the magnetized swing is 13–22% off.

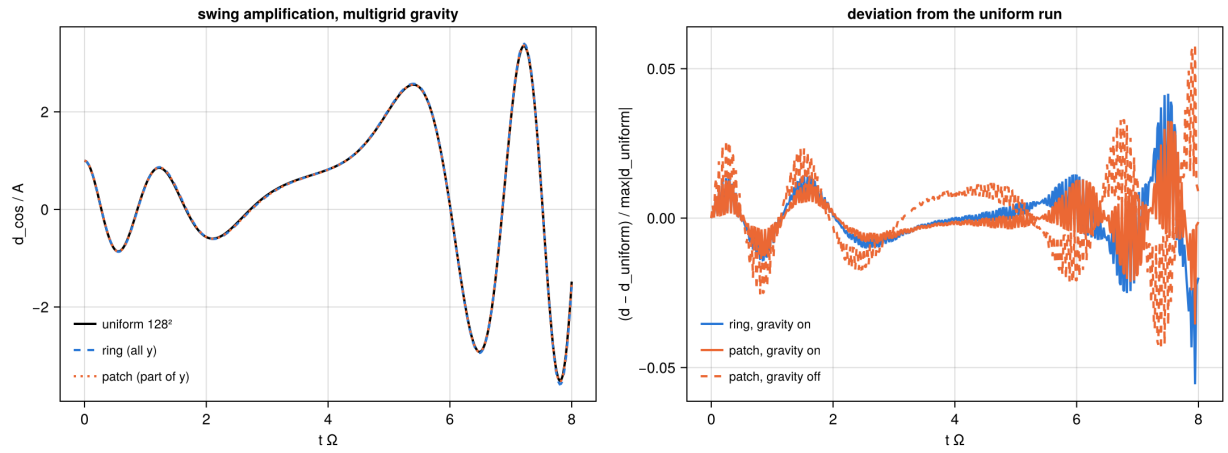


Figure 3: Swing amplification with multigrid self-gravity. Left: projection of the density onto the shearing wave, $d_{\cos}/A$, for the uniform, ring and patch meshes (indistinguishable at this scale). Right: deviation from the uniform run, with gravity (solid) and, for the patch, without (dashed). The fast oscillation of the deviations is present in the ring run as well. Script `figures/plot_fargo_amr_tests.jl`.

9.1 Regression

Uniform MHD with orbital advection (`mhd_compress_shwave`) is bit-identical to the parent code in the history and all VTK dumps: without refinement the remap's EMFs are not corrected and the arithmetic is unchanged.

9.2 The refined-mesh field error disappears with orbital advection

The three-minute reproducer of the Phase-4 field error (`inputs/tests/swing_mhd_repro.athinput`: $64^2 \times 4$ root cells, $8^2 \times 4$ blocks, $B_0 = 1$ along y, $\beta = 1.2$, no gravity, static level-1 ring $|x| < \pi/4$), run with and without orbital advection. Deviation over $t \leq 6$, on common times:

	vs linear theory		ring vs uniform twin
run	δ	c_y	c_y
uniform, no orbital advection	0.068	0.075	—
ring, no orbital advection	0.075	0.226	0.220
uniform, orbital advection	0.091	0.106	—
ring, orbital advection	0.062	0.074	0.033

With orbital advection the ring is closer to linear theory than its uniform twin, as refinement should be, and within 3.3% of it (δ: 3.0%), where the same ring without orbital advection is 22% off (Fig. 4, left). The two uniform runs agree with each other to 3%. So the Phase-4 defect needs the background shear flow carried in the state across the level interfaces; with orbital advection the state holds only the perturbation and the defect does not arise. (Comparing the twins row by row of the history, as `repro.py` did, is wrong when one run's step exceeds the history interval — the uniform run with orbital advection takes $\Delta t = 0.021$ against an output interval of 0.02 — and gave a spurious 52%; `repro.py` now interpolates to common times.)

9.3 Shared faces stay consistent

The history columns `bmis`/`bnrm` measure how far the coarse copy of each face shared with a finer block is from the mean of the fine copies:

mesh	copy mismatch / $\lvert B_n\rvert$ EMF correction off	on	max $\lvert\nabla\cdot B\rvert/\lvert B\rvert$
ring (level jumps at x_1 faces)	$\sim 10^{-2}$ (up to 0.7)	$\sim 10^{-15}$	4×10^{-15}
ring + half-height in z (x_1 and x_3 jumps)	median 1.8×10^{-2}	$\sim 10^{-15}$	3×10^{-15}
moving ring, AMR (280 blocks created, 224 destroyed)	median 3×10^{-3}	$\sim 10^{-15}$	5×10^{-15}
ring without orbital advection (stage EMFs only)	$\sim 10^{-15}$		4×10^{-15}

Without the correction the two copies drift apart by about 1% of the field (the ratio peaks where B_n passes through zero); each block stays divergence-free, which is why the history's $\nabla\cdot B$ does not see it. With it they agree to round-off, as the stage's own EMF correction keeps them without orbital advection (Fig. 4, right). The effect on the smooth swing wave is small (2–4×10^{-4} in δ and c_y), but a shared face with two different values is a monopole of the composite mesh. On 4 MPI ranks every field dump of the x_3-jump and AMR runs is bit-identical to the serial run.

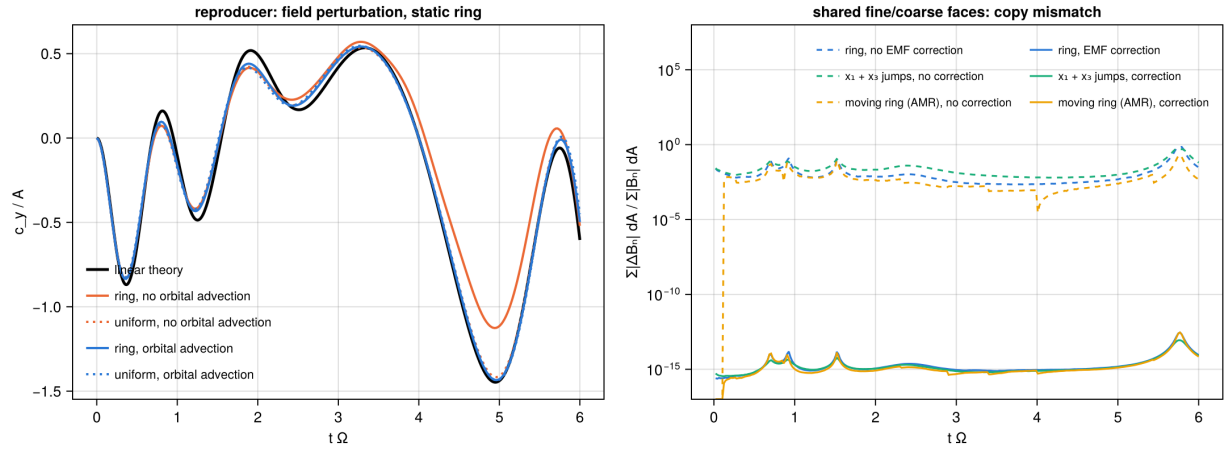


Figure 4: MHD on rings. Left: azimuthal field perturbation c_y of the refined-MHD reproducer against linear theory; without orbital advection the ring misses the swing's minimum (the Phase-4 defect), with it the ring follows its uniform twin. Right: mismatch of the two copies of the faces shared by fine and coarse blocks, with (solid) and without (dashed) the EMF correction of the face-field remap. Script `figures/plot_fargo_amr_mhd.jl`.

9.4 Magnetized swing amplification with gravity

The full Phase-4 test (`inputs/tests/swing_mhd.athinput`: $128^2\times16$ root cells, 16^3 blocks, multigrid gravity with $4\pi G\rho_0 = 1.9$, HLLD, PLM) on the two-column ring and the moving ring of Phase 4, now with orbital advection, against the uniform runs with orbital advection (`uF_b*`). Deviation from the uniform twin over $t \le 6$, with the Phase-4 values without orbital advection for comparison:

case	orbital advection (this work) δ	c_y	without (Phase 4) δ	c_y
static ring, $B_0 = 0.5$	0.035	0.017	0.025	0.071
static ring, $B_0 = 1.0$	0.014	0.014	0.039	0.133
moving ring, $B_0 = 1.0$	0.013	0.013	0.040	0.150

With orbital advection the refined runs follow their uniform twins to 1–4% in δ and 1–2% in c_y, against 7–15% in c_y without it. Against linear theory they are closer than the uniform runs, as refinement should be: c_y is off by 3.2% and 2.7% on the static rings ($B_0 = 0.5$, 1.0) and 2.9% on the moving ring, against 4.9% and 4.0% for the uniform runs. The shared faces stay consistent to round-off (copy mismatch $\leq 10^{-12}$ of $|B_n|$) and $\max|\nabla{\cdot}B|/|B| \leq 7 \times 10^{-15}$. The static rings (176 MeshBlocks) ran to $t = 10$; to $t \leq 8$ their c_y deviation is 1.7% and 2.6%. The moving ring (64–176 MeshBlocks; 448 created, 392 destroyed) also ran to $t = 10$: 1.9% and 2.3% to $t \leq 8$, with $\nabla{\cdot}B \leq 5 \times 10^{-15}$ throughout. All on 3 MPI ranks; inputs and runs in `validation/run/fargo_amr/mhd/swing/`.

So with orbital advection, refined MHD shearing boxes are validated on rings, with and without self-gravity; the refined-mesh field error of Phase 4 remains a property of the full-shear-flow frame.

9.5 Across x_2 level jumps

In the second step the face-field remap takes x_2 neighbours one level coarser or finer (implementation document), and MHD with orbital advection follows the hydro policy: `oa_annular` is optional.

Regression. Uniform MHD with orbital advection is still bit-identical to the parent code, and the ring, x_3-jump and moving-ring reproducer runs of the first step are bit-identical to their first-step results: the same-level path is unchanged.

Reproducer on a patch. The reproducer with the ring cut to four of eight block rows in y ($|x| < \pi/4$, $|y| < \pi/2$; 120 MeshBlocks), and with a moving box under AMR (the moving ring of the first step with an x_2 window $y_c = \sin t + 0.5\,t$, half-width 0.6; 64–106 MeshBlocks, 147 created, 126 destroyed):

mesh	vs uniform twin δ	c_y	copy mismatch / $\lvert B_n\rvert$ correction on / off	$\max\lvert\nabla{\cdot}B\rvert/\lvert B\rvert$
ring (first step)	0.030	0.033	10^{-15} / 10^{-2}	4×10^{-15}
static patch (x_2 jumps)	0.025	0.026	2×10^{-15} / 6×10^{-7}	4×10^{-15}
moving box, AMR	0.023	0.023	10^{-15} / 2×10^{-7}	4×10^{-12} (see below)

On the patch the field perturbation follows its uniform twin as closely as on the ring, and every shared face, now including the x_2 faces, stays consistent to round-off. (Without the correction the normalized mismatch is smaller than on the ring only because the x_2 faces add their normal field $B_y \approx B_0 = 1$ to the normalization; the absolute mismatch is of the same order.) On 4 MPI ranks every field dump of both runs is bit-identical to the serial run.

A pre-existing $\nabla{\cdot}B$ defect at AMR derefinement. In the moving-box run $|\nabla{\cdot}B|/|B|$ grows in steps, 10^{-13} to 5×10^{-12}, at the regrid events where blocks are *derefined*, and stays constant in between (so the remap and its correction conserve it). Two controls place it outside this work: the same moving box with $q = 0$ (the remap then shifts nothing) and with $q = 0$ and orbital advection off give the same jumps at the same events (3.9×10^{-13} at $t = 0.28$, 7.1×10^{-12} by

$t = 3$). The moving ring of the first step derefines too (392 blocks destroyed in the full swing run) and stays at 5×10^{-15}; a box adds fine neighbours across x_2 faces and x_1x_2 edges to a derefined block. The suspect is AthenaK's post-AMR face-field handling (`MeshRefinement::RepairAMRFC`, which recomputes the interior faces of created *and* derefined blocks, or the face ownership of the receives at those neighbours). It is small now but accumulates with every derefinement; it is left for a separate fix.

Magnetized swing with gravity on a patch. The full test of §9 (multigrid gravity, $128^2 \times 16$, 3 MPI ranks) with the two-column ring cut to half the azimuth ($|y| < \pi/2$):

	vs uniform twin, $t \le 6$		vs theory, $t \le 6$	
case	δ	c_y	c_y	$\max \lvert\nabla\cdot B\rvert/\lvert B\rvert$
patch, $B_0 = 0.5$	0.030	0.015	0.035	5×10^{-15}
patch, $B_0 = 1.0$	0.012	0.011	0.030	6×10^{-15}
ring, $B_0 = 0.5$ (above)	0.035	0.017	0.032	6×10^{-15}
ring, $B_0 = 1.0$ (above)	0.014	0.014	0.027	7×10^{-15}

The patch (120 MeshBlocks) follows its uniform twin as closely as the ring, 1–3% in δ and c_y, also to $t \le 8$ (1.4%, 1.7% at $B_0 = 1.0$), and linear theory better than the uniform runs (c_y: 3.5% and 3.0% against 4.9% and 4.0%). Shared faces agree to 5×10^{-15} of $|B_n|$, and $\nabla\cdot B$ stays at round-off. Runs in `validation/run/fargo_amr/mhd/swing/` (`ypF_b*`, input `ypatch.athinput`) and `mhd/stepB/` (reproducer).

10 Not yet validated

- **SC14 production.** The clump-following AMR of the gravito-turbulent box (`gravito_turb_l64z16_clump_amr_overlay.athinput`) restarted from an archived $\beta = 3$ state, once with `oa_annular = true` and once without: block count, cost per cycle and whether the timestep limit binds. Needs a cluster.
- **GPU.** The kernels follow the existing team-scratch pattern but have run only on the CPU (serial and MPI).
- **$\nabla\cdot B$ at AMR derefinement** next to fine x_2 neighbours (§9.5): pre-existing in AthenaK, 10^{-13}–10^{-12} per event, to be fixed separately. Production MHD runs with box-shaped AMR should watch the history's `divb`.
- **Non-FARGO refinement with x_2 level jumps** fails in the parent code (§6); not addressed here.

11 Reproduction

```
cd validation/run/fargo_amr        # README.md: one command per run directory
python3 compare.py unif128 unif ring ypatch amrring amrbox   # shwave table
python3 errmap.py 00010                                      # error at the level jumps
(cd swing && python3 cmp.py)                                 # swing table
python3 ../swing_mhd_repro/repro.py 6 base fargo fargo_nofc  # MHD reproducer pairs
cd ../.. && julia figures/plot_fargo_amr_shwave.jl           # Fig. 2
julia figures/plot_fargo_amr_tests.jl                        # Figs. 1 and 3
julia figures/plot_fargo_amr_mhd.jl                          # Fig. 4
```